Things that start with “ph”

LING/PSYC 406 Session 1.2
(Covering Menn, Chapter 1A)

Agenda for this lecture/discussion session

1. Mispronouncing contest

2. The logic of the IPA and feature-based descriptions of spee
sounds

3. Minimal pairs, allophones and the descriptive power of the
phoneme

4. Some demonstrations of how messy this stuff can be in re

Mispronouncing contest

When it's your turn:

1. Pronounce your word correctly
2. Mispronounce it clearly and loudly for th
class

When it's not your turn, score your classma
mispronunciations on a scale of 1-10.


It is so delicious with tzatziki


BARTENURA

[your examples]

Mispronouncing contest

OK, that was fun. Now some questions:

1. How did you come up with your mispronunciation?

2. What criteria did you use to score your classmates?

The logic of the IPA and feature-bas descriptions of speech sounds

By now you've understood something important about alphabetic writing system *are not phonemes*.

English is especially ambiguous (as Menn puts it later, *irregular*), but even less irr systems pose problems when we make comparisons across languages.

For example, Pinyin is an (relatively recently) invented system for representing M It is perfectly regular. Try pronouncing this phrase from an elementary Mandarin textbook:

Wo qu tushuguan. （我去图书馆）

French is also very regular. Try pronouncing this simple sentence in French:

Comment tu t'appelles?

The logic of the IPA and feature-bas descriptions of speech sounds

So, we need a standard system for representing the sounds of the wor languages in a way that accurately captures their similarities and diffe

Here's the chart of IPA symbols for consonants:

CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngea
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ	
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ	
Trill	ʙ			r					ʀ	
Tap or Flap		ⱱ		ɾ		ɽ				
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ
Lateral fricative				ɬ ɮ						
Approximant		ʋ		ɹ		ɻ	j	ɰ		

Note this chart is organized by *features:*

PLACE (x-axis: front to back of vocal tract)
MANNER (y-axis: degree of closure – closed to open)
VOICING (pairs: onset of voicing before/during or after closure)

CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngea
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ	
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ	
Trill	ʙ			r					ʀ	
Tap or Flap		ⱱ		ɾ		ɽ				
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ
Lateral fricative				ɬ ɮ						
Approximant		ʋ		ɹ		ɻ	j	ɰ		
Lateral approximant				l		ɭ	ʎ	ʟ		

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged im

/p/
+ plosive
+ bilabial
- voiced

/b/
+ plosive
+ bilabial
+ voiced

This allows us to describe the similarities and differences between sounds in a relatively precise and well-defined way:

/p/ and /b/ both have the same **manner** and **place of articulation**.

But /p/ and /b/ differ in the timing of phonation relative to the release of the closure **(voicing)**.

This difference is analogous to the difference between /t/ and /d/.

So we can talk in general about “the voice

[pʰ]

+ plosive
+ bilabial
- voiced
+ aspirated

In fact, the way the contrast between “voiced” and
“unvoiced” stops is produced differs across
languages, and across contexts within a language

IPA also provides a compact way of representing
this.

[p]

+ plosive
+ bilabial
- voiced
- aspirated

In English, *word-initially,* we often use an ***unvoiced***
aspirated stop for the first sound in “peer” and an
unvoiced, unaspirated stop for the first sound in
“beer.”

Convince yourself of this by putting your hand in
front of your mouth and saying those words.

[b]

+ plosive
+ bilabial
+ voiced

In most varieties of French, *au contraire,* the contr
is really between [p] for “pou” and [b] “boue.”

/p/
+ plosive
+ bilabial
- voiced

/b/
+ plosive
+ bilabial
+ voiced

[pʰ]
+ plosive
+ bilabial
- voiced
+ aspirated

[p]
+ plosive
+ bilabial
- voiced
- aspirated

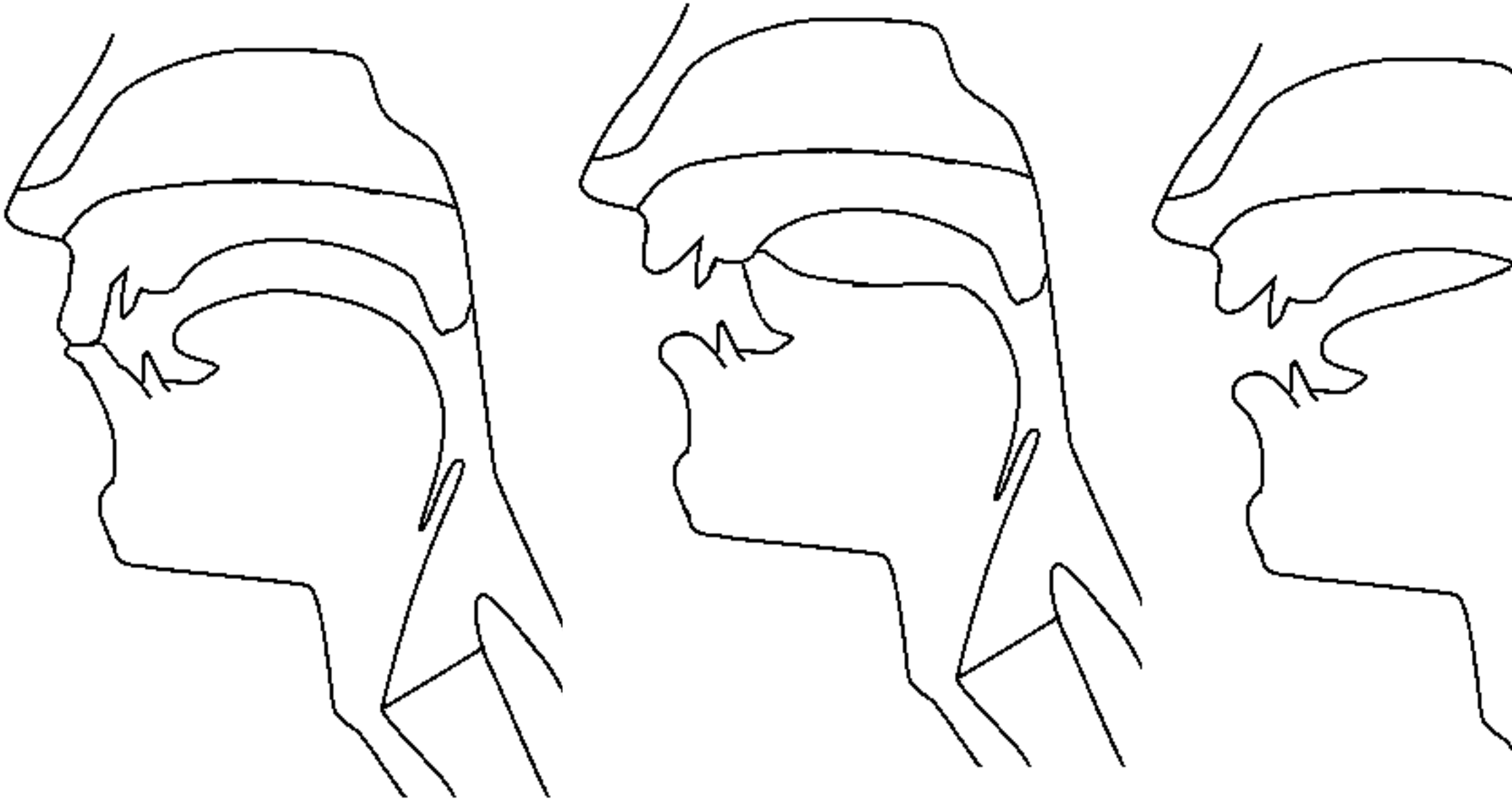
[b]
+ plosive
+ bilabial

You may have notice
I switched from slas
square brackets fro
slide to the next.

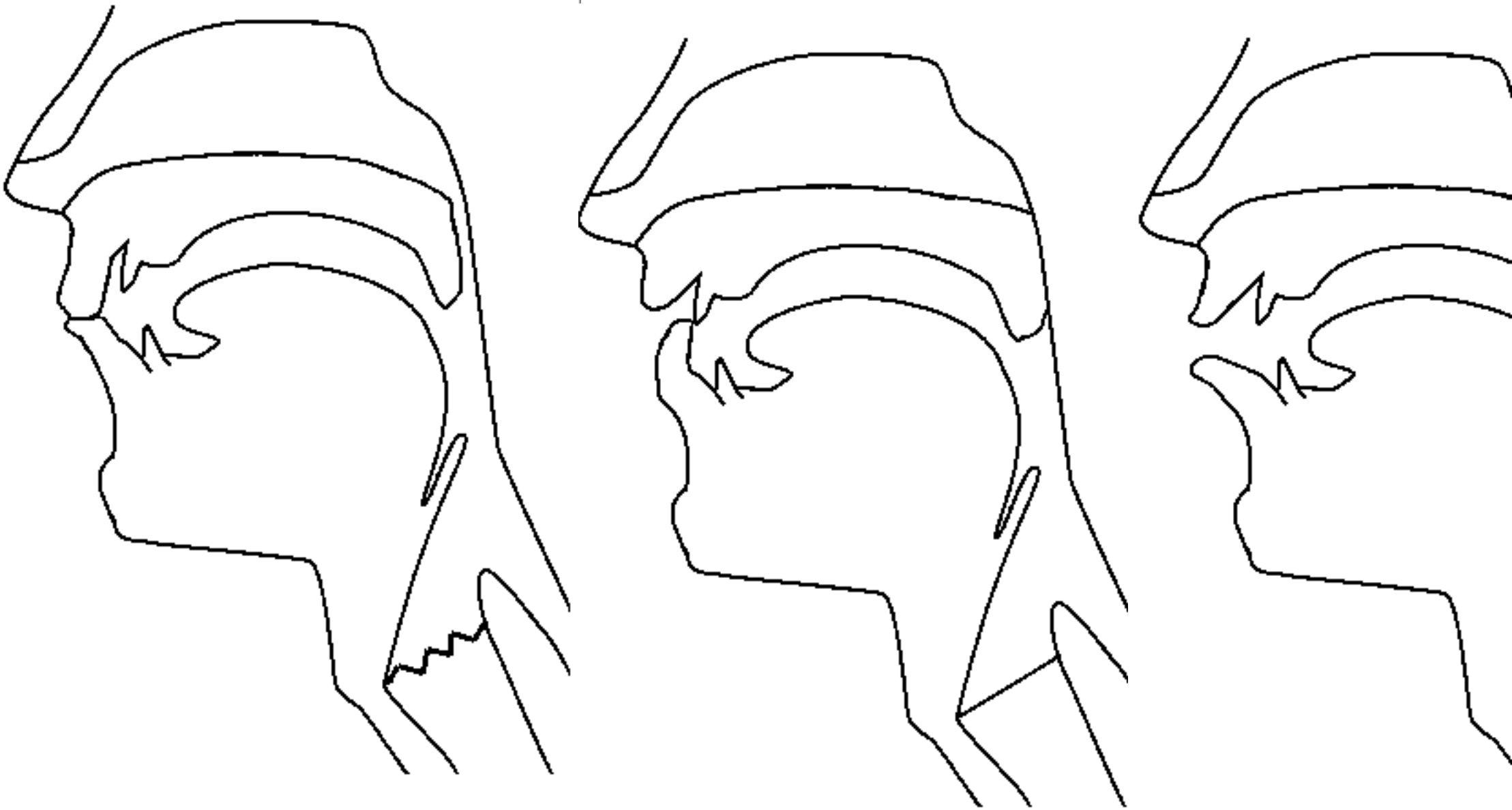
Slashes indicate *bro*
transcription, i.e.,
transcribing the *pho*
of an utterance.

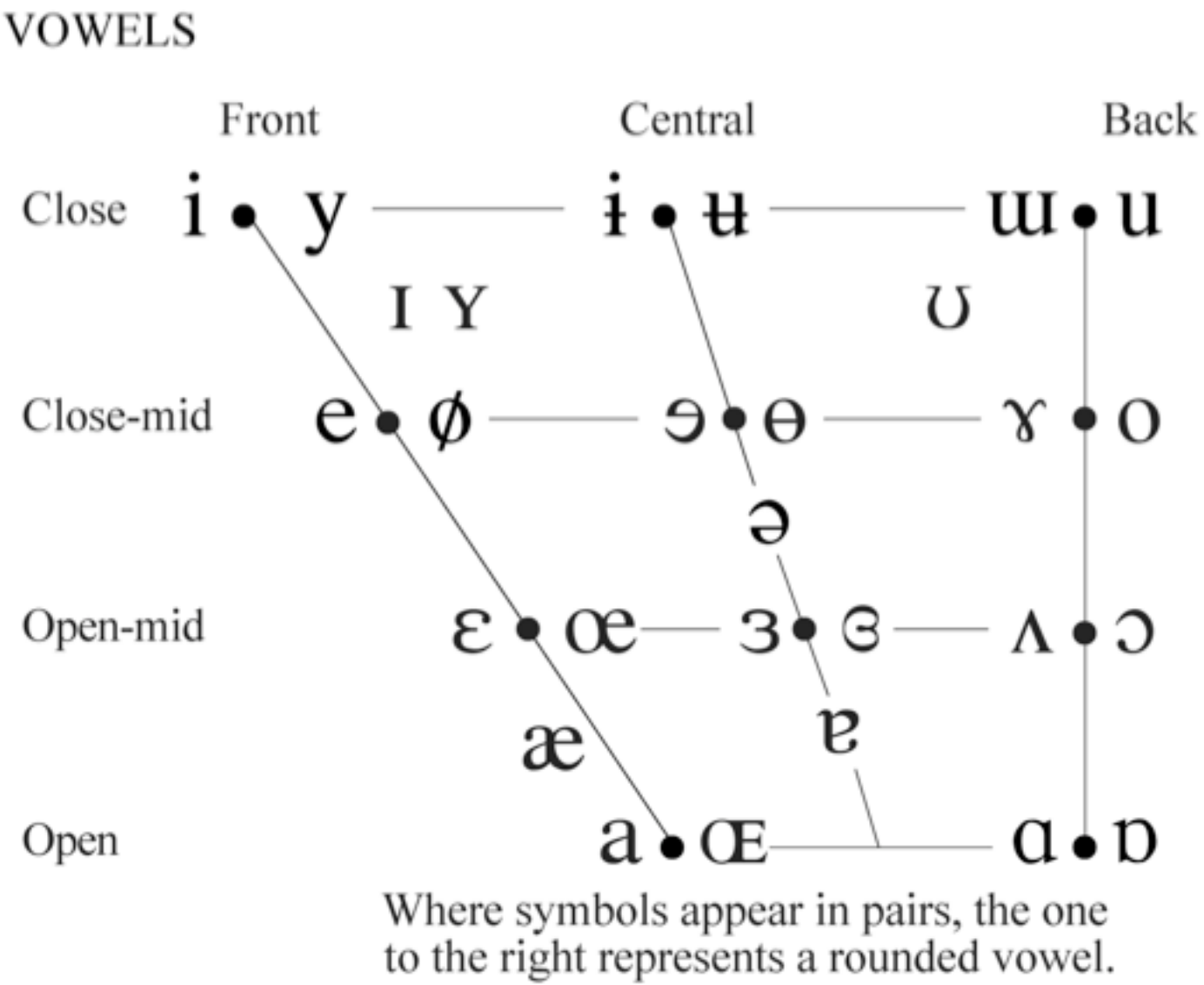
Brackets indicate *na*
transcription, i.e., inc
as much detail as po

Phonetic features, especially for consonant
anchored to how speech is produced



Phonetic features, especially for consonant anchored to how speech is produced





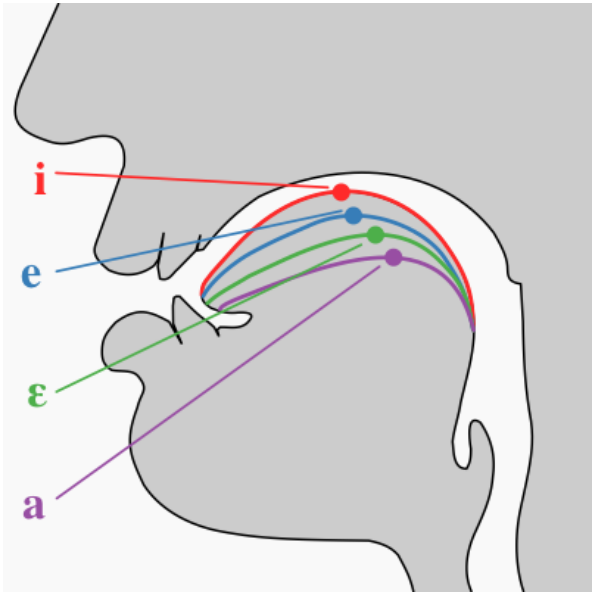
It gets more complicat
but:

“close” vowels tend to l
jaw position than “oper
this dimension is just c

and the tongue is gene
forward in the mouth f
“back” vowels.

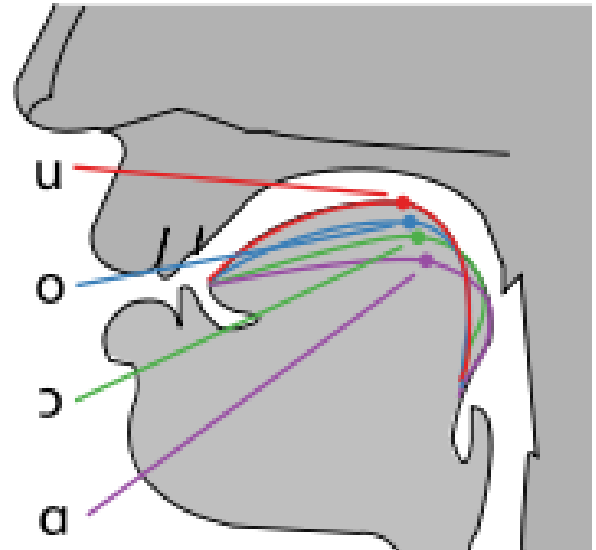
…and of course “round
rounding of the lips

Note /y/ is the vowel in
word “tu” and the Man
“qu/ 去 .” It sounds like
English speakers, but y
convince yourself othe
make an /i/ and chang



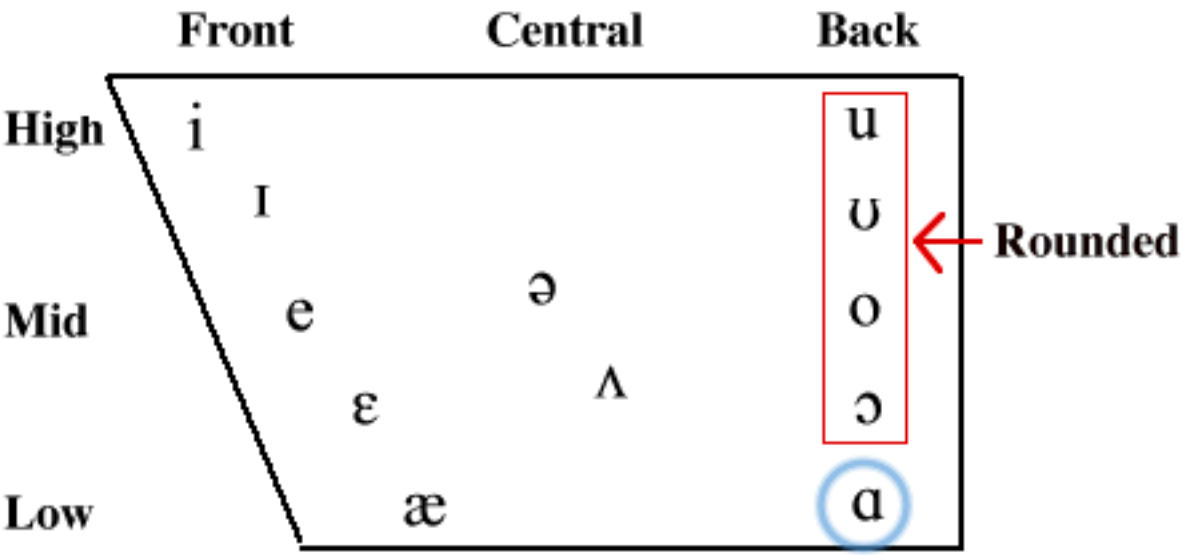
The dot indicates the center o
the tongue body for each
vowel.

But, as usual, things are more
complicated:



Some vowels, like the
diphthongs in “boat” and “bait
are very dynamic.

And how vowels are produced

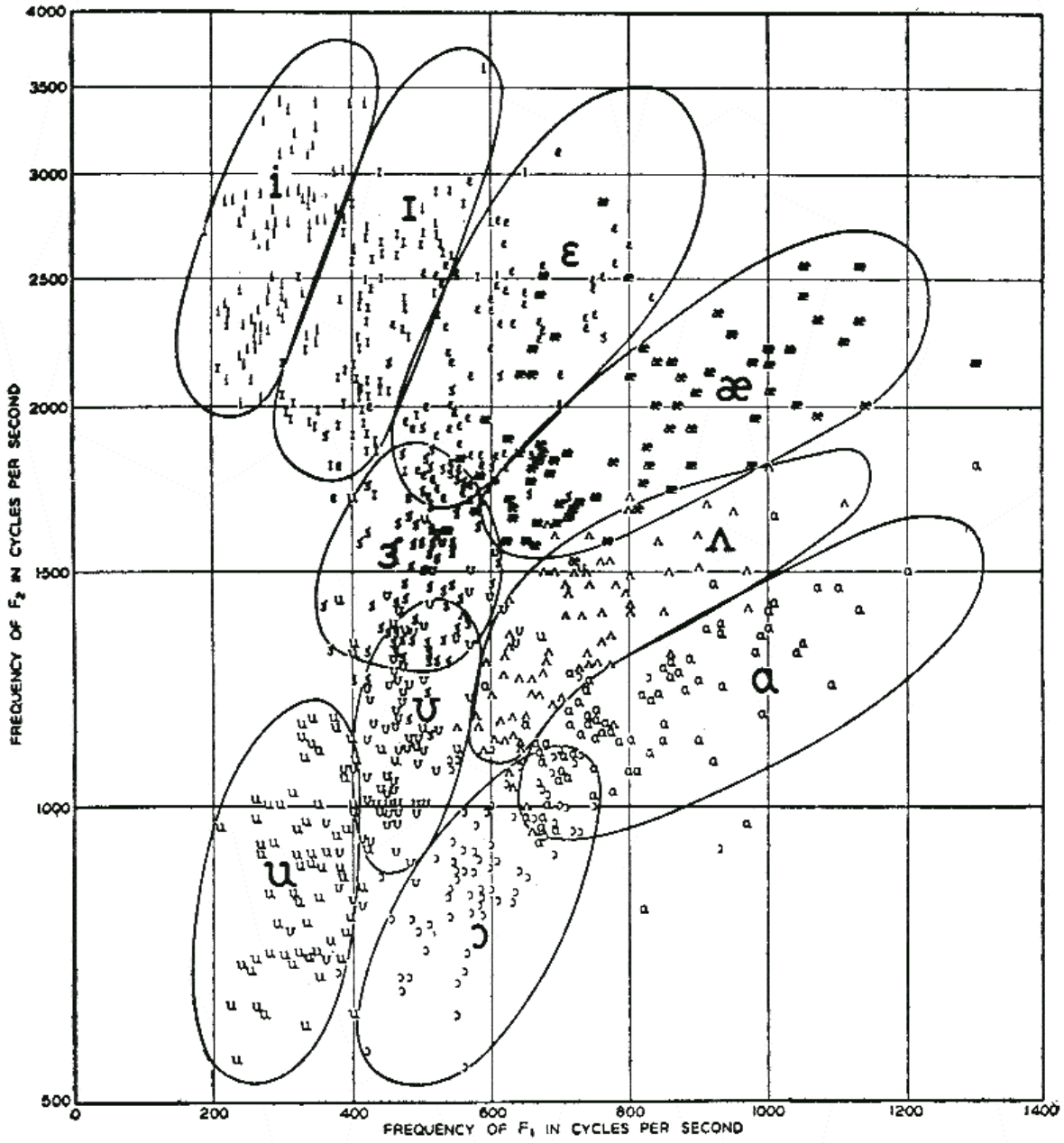


Dialect variability:

Even within American vowels are very salien different across diale

In my dialect (mid-Atl more specifically oute boroughs of NYC) I di between the two vow lower right hand corn most people from oth of the country do not

So, CAUGHT and COT *homophones* for me.



Frequency of second formant *versus* frequency of first

The fact that

I understand people
"Aaron" and "Erin" w
same vowel,

and that you have h
producing these diff
vowels for a couple
now, and may just b
it because I'm drawi
attention to it...

...is our first encoun
the dreaded **lack of**
invariance problem
come up again and
especially when we
speech.

Minimal Pairs

Consider how we established that I really do have two different low back vowels.

We had to find two different **words** in which the distinction between them depended on the difference between those vowels.



Minimal pairs are important for demonstrating **contrast**.

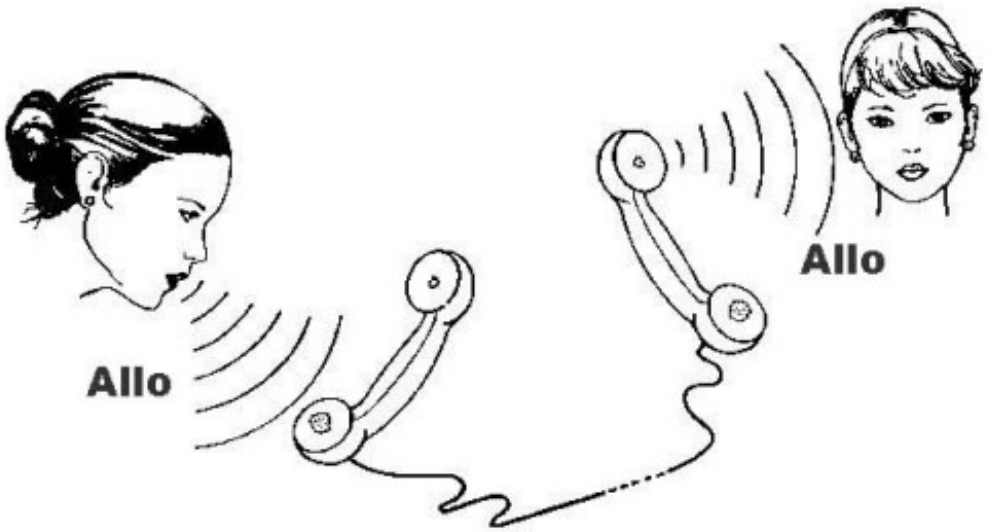
If two phones are contrastive -- i.e., you can find a minimal pair where the difference between them results in a difference in meaning -- they are examples of different **phonemes**.

Minimal pairs

Evidence that phonemic contrasts depend on a particul
feature also comes from minimal pairs.

For example, the pairs /b/ - /p/, /t/ - /d/ and /k/ - /g/ all c
on *voicing* in English. We know this from comparison of
minimal pairs in each language.

Allophones



We also saw that there is a ton of v
phonemes are realized.

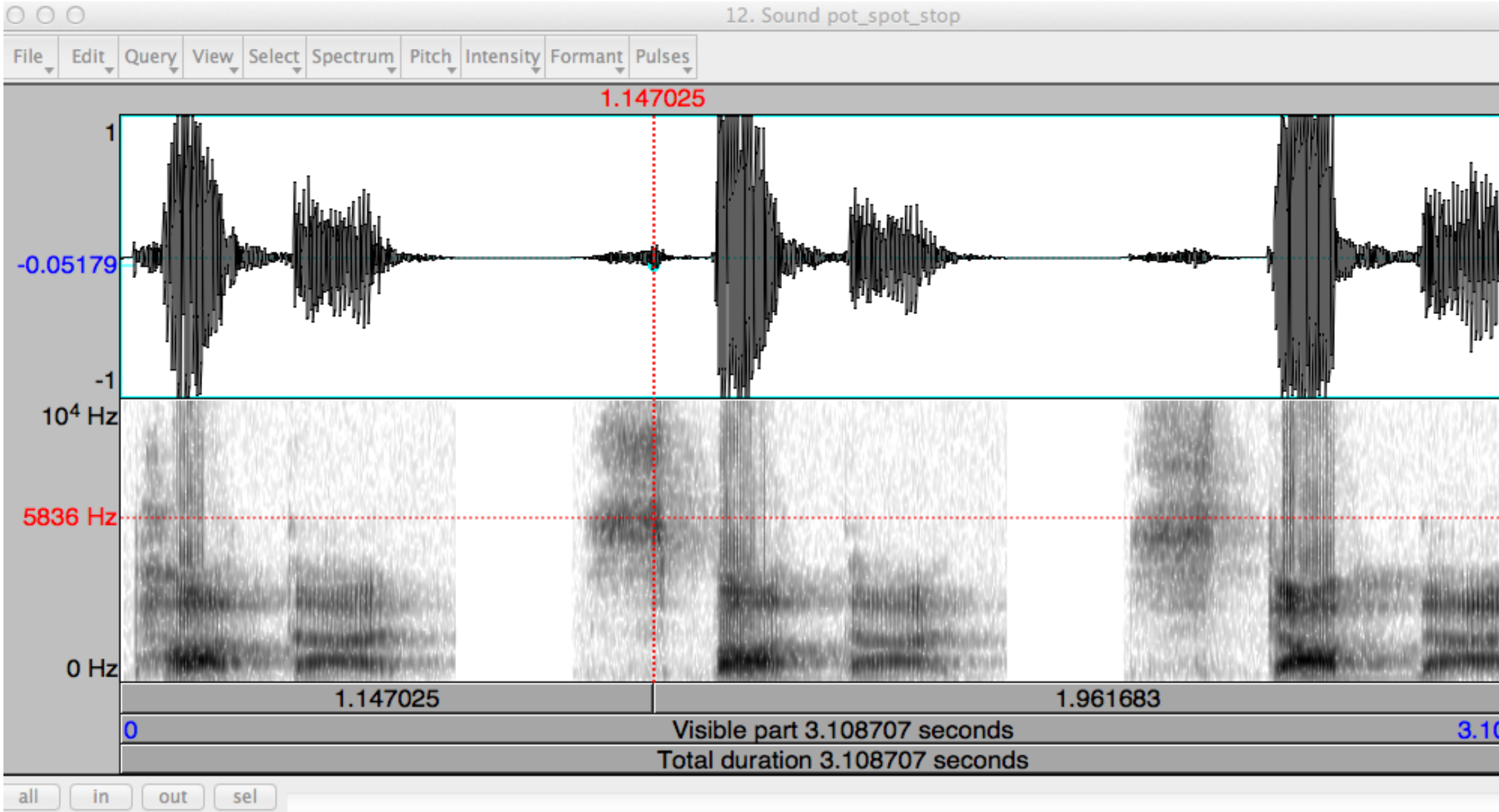
Some of this variability is very pred
can talk about the different **phones**
different realizations of the same **p**

These are called **allophones**.

Let's use Praat to make recordings
phrases:

"pot that" .... "spot that" ... "s

Allophones of /p/



Allophones

Now.

We do sometimes say [stɑpʰ], often when stressing this word at the a sentence, or trying to speak clearly (in “citation form”).

But there is no difference in meaning between [stɑp̚] and [stɑpʰ].

In fact **we can’t find a minimal pair** for these two phones *in these phonological contexts* in English, so we call them allophones (allo = different, so “different *phones* for the same *phoneme*.”)

Another interesting point is that these different forms alternate in different morphological forms, consider: “stop,” “stops,” “stopped,” “stopping” -- when do you release and aspirate the /p/?

A brief clarification:

At this point, you may be confused about what “voicing” is.

This is because it gets used in two different ways:

- When we are doing *narrow transcription*, and describ
specific *phones*, we call something “voiced” if there’s vibrati
the vocal folds during the whole gesture/sound.

- In *broad transcription*, i.e., describing *phonemes,* we c
say that voicing is *realized* in a number of different ways,
including things like presence or absence of aspiration at th
beginnings of words, or relative duration of voicing to closu
in “writer” vs. “rider.”

Syllables



[spɑroʊ] or [zbɑroʊ] but never [sbɑroʊ] whatever you think you're saying…

The discussion of allophones leads n
point about **phonotactics**, roughly, w
phonemes can occur together in syll
structures.

We can see this clearly with *clusters* l
can't occur in English words.

When forced to deal with phonotacti
clusters, e.g. in foreign names like Zb
(Polish) or Gbagbo (Kru), we tend to
and say [zɨ'bIg.nʲu] or [gə.bɑ.gə.boʊ]

Rules about syllable structure turn o
complex, but we'll see in a few weeks

Summary so far:

Phonemes are *not* letters.

In fact, they're not even specific *sounds* or *gestures*, they're "the smallest phonetic unit in a language that is capable of conveying distinction in meaning."

We've seen that this can be complicated in practice.

The International Phonetic Alphabet gives speech scientists a wa communicating about the similarities and differences between s across languages.

So it's easy to think of the IPA as an "alphabet of phonemes."

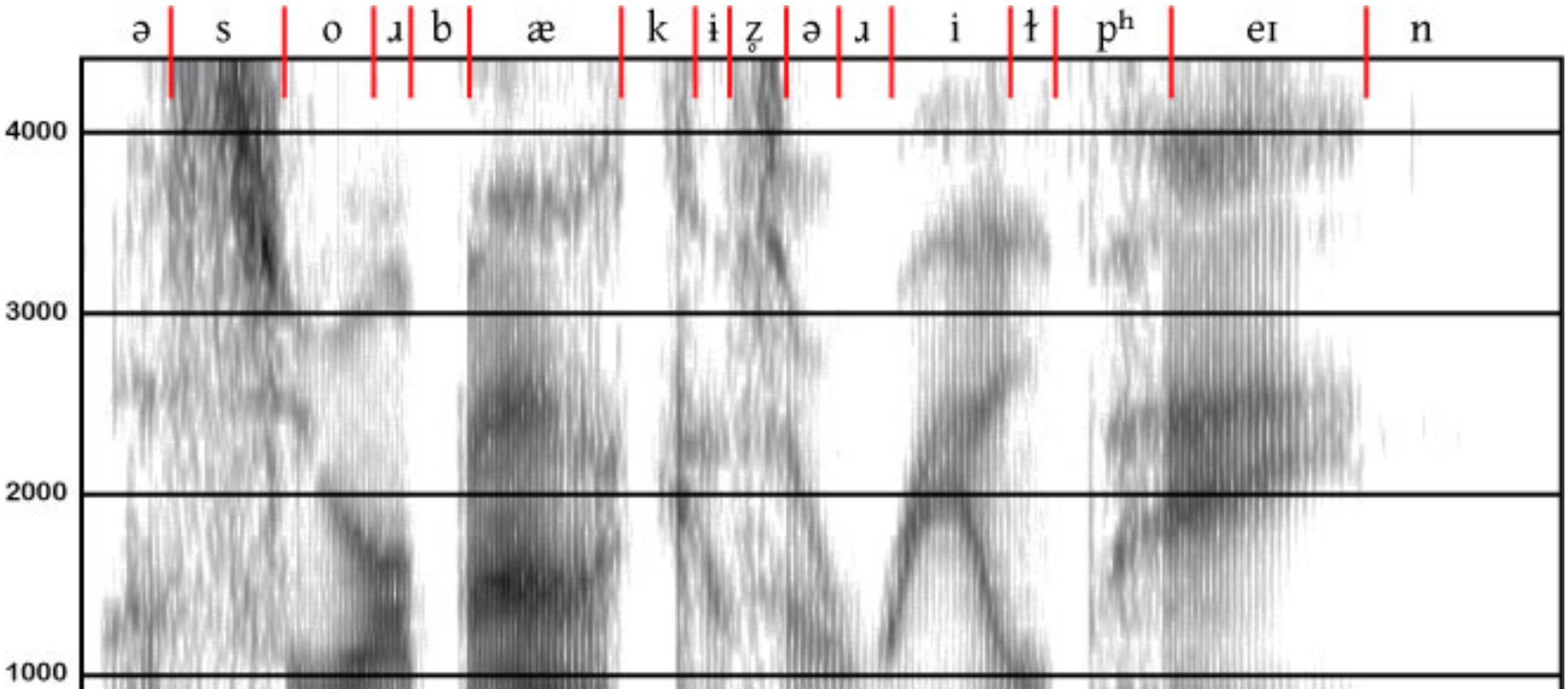
But this would be a mistake because *phonemes are not letters*.

Why do I keep insisting that *phonemes are no*

It's easy for literate people to imagine that speech is produced
orderly stream of letter-like units.

But it is nothing like that.

Here's a spectrogram of the sentence: "A sore back is a real pai



Here’s what the articulators look like in action:



“The best analogy I have been able to think of for this is a very homely one. Imagine a row of Easter eggs carried along a moving belt; the eggs are of various sizes, and variously colored, but not boiled. At a certain point, the belt carries the row of eggs between the two rollers of a wringer, which quite effectively smash them and rub them more or less into each other. The flow of eggs before the wringer represents the series of impulses from the phoneme source; the mess that emerges from the wringer represents the output of the speech transmitter. At a subsequent point, we have an inspector whose task it is to examine the passing mess and decide, on the basis of the broken and unbroken yolks, the variously spread-out albumen, and the variously colored bits of shell, the nature of the flow of eggs which previously arrived at the wringer. Note that he does not have to try to put the eggs together again—a manifest physical impossibility—but only to identify.

The inspector represents the hearer.”

Hockett (1955)

In fact, although it goes beyond the scope of this course, you should know that people have started to question whether are "real," either psychologically or linguistically...

Lotto, Andrew J. and Holt, Lori L., "The Illusion of the Phoneme" (2000). *Department of Psychology*. Paper 16
http://repository.cmu.edu/psychology/166

Child Development, January/February 2000, Volume 71, Number 1, Pages 240–249

The Ontogeny of Phonological Categories and the Primacy of Lexical Learning in Linguistic Development

Mary E. Beckman and Jan Edwards

ELSEVIER

NEW IDEAS IN PSYCHOLOGY

New Ideas in Psychology 25 (2007) 143–170
www.elsevier.com/locate/newideapsych

How are words stored in memory? Beyond phones and phonemes

Robert Port

Department of Linguistics, Department of Cognitive Science, Indiana University, USA

Available online at www.sciencedirect.com
SCIENCE DIRECT®

Journal of Phonetics 31 (2003) 305–320

Puzzle-solving science: the quixotic quest speech perception

Stephen D. Goldinger[a,*], Tamiko Azun

[a] *Department of Psychology, Arizona State University, Box 871104, Tempe,*
[b] *Department of Speech & Hearing Science, Arizona State Unive*

Received 15 August 2002; received in revised form 18 March 2003; accep

For Thursday :

Read: Menn Chapter 1

Prepare: Make up a word. Try to come up with a w
whose meaning would be obvious to someone wh
never heard it before.

Extra stuff:

How to read a spectrogram

Sounds of the World's Language s online

Maps and quizzes based o Harvard Dialect Survey (fo English): https://mydialect.

Note that by leaving out ag ethnicity, SES, etc., these ki of maps tend to paper ove lot of variability.